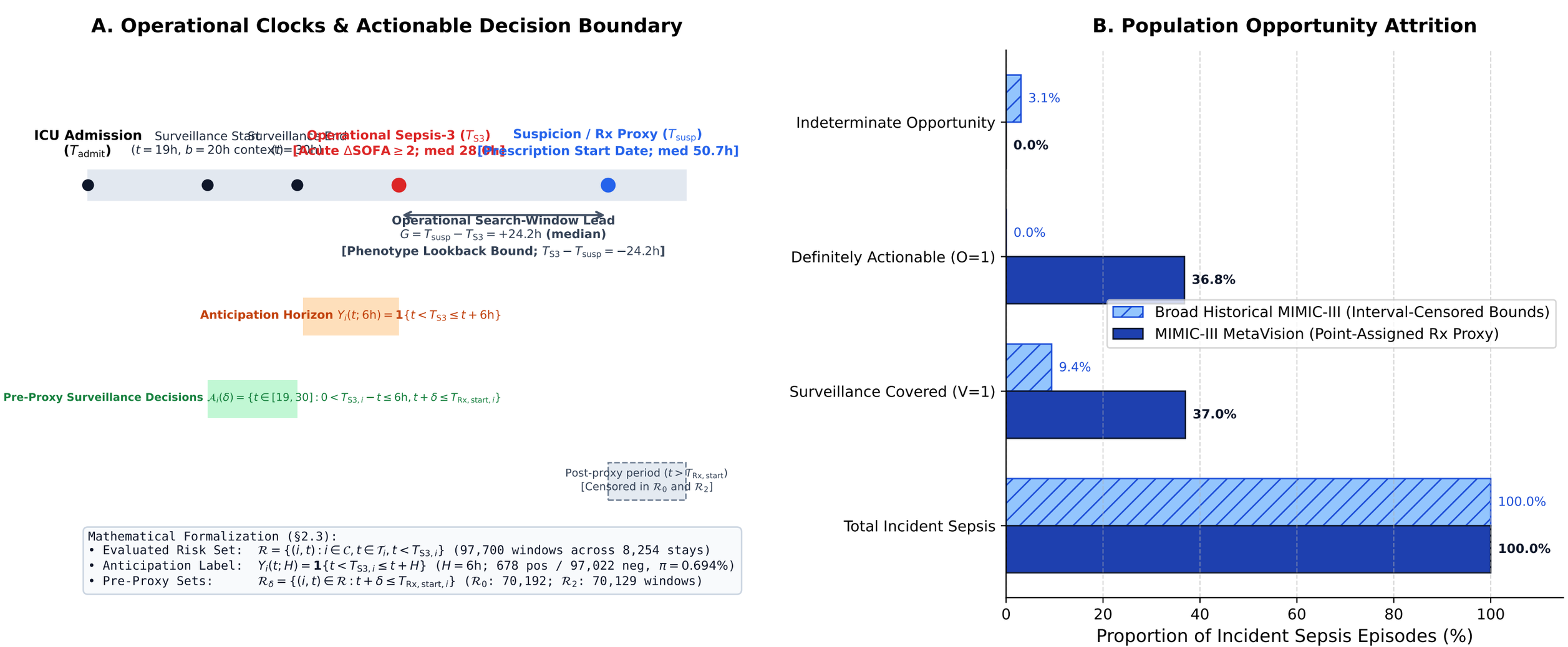

A. Operational Clocks & Actionable Decision Boundary
ICU Admission ($T_{admit}$)
Surveillance Start ($t = 19$h, $b = 20$h context)
Surveillance End ($t = 30$h)
Operational Sepsis-3 ($T_{S3}$)
[Acute ΔSOFA ≥ 2; med 28.0h]
Suspicion / Rx Proxy ($T_{susp}$)
[Prescription Start Date; med 50.7h]
Operational Search-Window Lead
$G = T_{susp} - T_{S3} = +24.2$h (median)
[Phenotype Lookback Bound; $T_{S3} - T_{susp} = -24.2$h]
Anticipation Horizon $Y_i(t; 6h) = \mathbf{1}\{t < T_{S3} \le t + 6h\}$
Pre-Proxy Surveillance Decisions $\mathcal{A}_i(\delta) = \{t \in [19, 30] : 0 < T_{S3,i} - t \le 6h, t + \delta \le T_{Rx,start,i}\}$
Post-proxy period ($t > T_{Rx,start}$)
[Censored in $\mathcal{R}_0$ and $\mathcal{R}_2$]
Mathematical Formalization (§2.3):
• Evaluated Risk Set: $\mathcal{R} = \{(i,t) : i \in \mathcal{C}, t \in \mathcal{T}_i, t < T_{S3,i}\}$ (97,700 windows across 8,254 stays)
• Anticipation Label: $Y_i(t; H) = \mathbf{1}\{t < T_{S3,i} \le t + H\}$ ($H = 6$h; 678 pos / 97,022 neg, $\pi = 0.694\%$)
• Pre-Proxy Sets: $\mathcal{R}_\delta = \{(i,t) \in \mathcal{R} : t + \delta \le T_{Rx,start,i}\}$ ($\mathcal{R}_0$: 70,192; $\mathcal{R}_2$: 70,129 windows)
B. Population Opportunity Attrition
Indeterminate Opportunity
Definitely Actionable (O=1)
Surveillance Covered (V=1)
Total Incident Sepsis
3.1%
0.0%
0.0%
36.8%
9.4%
37.0%
100.0%
100.0%
Broad Historical MIMIC-III (Interval-Censored Bounds)
MIMIC-III MetaVision (Point-Assigned Rx Proxy)
0
20
40
60
80
100
Proportion of Incident Sepsis Episodes (%)